One of the main data-quality issues encountered during the project was the difference in temporal coverage between the Maddison Project and the World Development Indicators (WDI). Maddison provides historical data from 1900 to 2022, whereas WDI indicators generally cover a shorter period, mostly starting around 1960. These differences also affected our report: we opted to use GNI per capita instead of GDP per capita for the analyses (as it reflects better the income received by a certain country), meaning that the corresponding comparisons only reflect data from 1960 onwards.

Another significant issue was the presence of missing values, even within the expected coverage period of an indicator. For example, Maddison data for countries outside the Focus and Western Europe groups was often unavailable during the earliest years. Similarly, in WDI, GNI per capita data began some years later than life expectancy data. These gaps limited the number of countries and years that could be included in certain analyses and made direct comparisons more difficult.

The transformation of WDI data also presented challenges during ETL development. The indicator codes used in the original dataset were difficult to interpret, making it harder to identify the meaning of each indicator and correctly transform the data from long to wide format. This increased the complexity of developing the ETL pipeline and required additional effort to map indicators to the appropriate columns in the dimensional model.

Finally, we decided not to include the European Union (EU) aggregate in our analyses because Maddison did not provide the necessary data for this entity. Although WDI contained EU data, the lack of corresponding Maddison measures prevented us from integrating it consistently into our comparisons. This decision limited the potential scope of our analysis, as including the EU could have provided an additional regional perspective.